Mathematical Statistics II

Point Estimation: Principles and Theory

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Contents

1 Introduction 1

2 Fisher's Information 1

3 Asymptotic properties of the MLE and Bayesian estimators 7

4 Bias, MSE, and Consistency 11

5 Sufficiency 20

1 Introduction

Introduction

- We now introduce principles of parameter estimation, as well as associated theory.
- The three approaches we have considered thus far (MoM, MLE, Bayes) are great starting points, but we would like to study general properties of parameter estimations, and how these approaches align with these properties.
- There is no "correct" order in which these material are presented, so we deviate a bit from our textbooks.
- This work is largely based on Rice (2007, Sections 8.5.2, 8.7–8.8), Casella and Berger (2024, Chapters 6, 7, 10), as well as some supplemental material from Pawitan (2001).

2 Fisher's Information

Curvature of the Likelihood Function

- Both the MLE and Bayesian estimates are forms of *Likelihood based inference*.
- For the MLE, we are interested in the maximum of the likelihood function.
- For Bayesian inference, the posterior distribution is the (scaled) product of the likelihood and a prior distribution:
$$\pi(\theta|x) \propto L(\theta) \times \pi(\theta).$$
- In both instances, the shape of the likelihood function $L(\theta)$ is of interest.

- *Coming up:* We'll introduce the *score* function, and *Fisher's Information*. The latter can be used to obtain uncertainty estimates for *any* parameter estimated using the MLE technique.

  Definition: The score function

  Let $L(\theta) = f(x^*; \theta)$ be the likelihood of some model, conditioned on the observed data (x^* is assumed to be multi-variate). Then we define the score function $S(\theta)$ to be:

$$S(\theta) = \frac{\partial}{\partial \theta} \log L(\theta).$$

- Assuming the MLE $\hat{\theta}$ exists and is not a boundary point, then:

$$S\big(\hat{\theta}\big) = 0.$$

- As written, the score function is dependent on a fixed data set: x^*.
- We could also expand this by considering it as a random function of any possible data X, which is random.
- Since we are interested in the curvature of the likelihood function, we should also consider the second derivative.
- At the MLE, the second derivative of the log-likelihood will be *negative*, so it's natural to consider the negative second derivative of the log-likelihood:

$$I(\theta) = -\frac{\partial^2}{\partial \theta^2} \log L(\theta).$$

- Again, we are treating the likelihood function $L(\theta)$ as a regular (non-random) function, fixed on x^* (this is slightly different than many texts).
- In this case, we call $I(\theta)$ the *observed Fisher's Information*, at the point θ.

  Comment: joint information

  We can define the information for just a single point $X_i = x_i^*$,

$$I_i(\theta) = -\frac{\partial^2}{\partial \theta^2} \log f_{X_i}(x_i^*; \theta),$$

  or for the entire collection: $X_{1:n} = x_{1:n}^*$:

$$I(\theta) = -\frac{\partial^2}{\partial \theta^2} \sum_i \log f_{X_i}(x_i^*; \theta) = \sum_i I_i(\theta).$$

- The curvature at the MLE is of particular interest.

  Definition: The Observed Fisher Information

  Let $L(\theta)$ be the likelihood function, fixed on the observed data x^*. Unless otherwise specified, the *observed Fisher information* is understood to be the negative second derivative of $\log L(\theta)$, evaluated at the MLE:

$$I\big(\hat{\theta}\big) = -\frac{\partial^2}{\partial \theta^2} \log L\big(\theta\big)\Big|_{\theta=\hat{\theta}}$$

  Note that when θ is multi-variate, this is a matrix with entries:

$$\Big[I\big(\hat{\theta}\big)\Big]_{i,j} = -\frac{\partial^2}{\partial \theta_i \partial \theta_j} \log L(\theta)\Big|_{\theta=\hat{\theta}},$$

  also denoted $I\big(\hat{\theta}\big) = -\nabla^2 \log L\big(\hat{\theta}\big)$.

- Note that large values of $I(\hat{\theta})$ in the univariate case (and large eigen-values / determinant in the multivariate case) indicate a "tight peak" around the maximum, suggesting less uncertainty about the parameter θ.

Observed Fisher's information, Binomial Distribution

Let $X \sim \text{Binom}(n, \theta)$. Find the MLE by setting the score function equal to zero, and then use that to compute the observed Fisher's information.

Proof. The likelihood function for the Binomial model is:

$$L(\theta) = f(X; \theta) = \binom{n}{X} \theta^X (1-\theta)^{n-X}.$$

The binomial coefficient in front is only a multiplicative constant, and doesn't change the overall shape or maximum of the likelihood function $L(\theta)$.

Taking the logarithm, we have:

$$\ell(\theta) = C + X \log(\theta) + (n - X) \log(1 - \theta).$$

The *score function* is the partial derivative of ℓ, with respect to θ:

$$S(\theta) = \frac{\partial}{\partial \theta} \ell(\theta) = \frac{X}{\theta} - \frac{n - X}{1 - \theta}.$$

Setting equal to zero, we can solve for the mle:

$$\hat{\theta} = \frac{X}{n}.$$

Now to compute the observed Fisher's information, we take the second derivative:

$$I(\theta) = -\frac{\partial^2}{\partial \theta^2} \ell(\theta) = \frac{X}{\theta^2} + \frac{n - X}{(1 - \theta)^2}$$

Plugging in the MLE $\hat{\theta} = X/n$, and some simplification gives:

$$I(\hat{\theta}) = \frac{n}{\hat{\theta}(1 - \hat{\theta})}.$$

□

Observed Fisher's information, Normal Distribution

Suppose $X_1, \ldots, X_n$ are iid from $N(\theta, \sigma^2)$, with known σ^2. Find the MLE by setting the score function to zero, and then calculate the joint information, $I_n(\theta)$ at the MLE.

Proof. Under the IID assumption, we have the likelihood function (joint distribution of the X, evaluated at x^*):

$$\begin{aligned} L(\theta) = f(x^*; \theta) &= \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} \exp\Big\{ -\frac{1}{2\sigma^2} \big(x_i^* - \theta\big)^2 \Big\} \\ &= C \exp\Big\{ -\frac{1}{2\sigma^2} \sum_{i=1}^{n} \big(x_i^* - \theta\big)^2 \Big\}. \end{aligned}$$

Here, we just note that C is some constant that does not depend on the parameter of interest, θ. Taking logarithms:

$$\ell(\theta) = C' - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\big(x_i^* - \theta\big)^2.$$

The score function is the partial derivative of $\ell(\theta)$, with respect to θ:

$$S(\theta) = \frac{\partial}{\partial\theta}\ell(\theta) = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i^* - \theta).$$

Setting equal to zero, solving for θ, results in the MLE:

$$\hat{\theta} = \bar{x}_n^*.$$

Taking the negative second derivative gives the observed Fisher information:

$$I(\theta) = \frac{n}{\sigma^2}.$$

Note that this is a constant with respect to θ, so we have:

$$I(\hat{\theta}) = I(\theta) = \frac{n}{\sigma^2}.$$

Note: In this example, note the relationship between $I(\hat{\theta})$ and $\mathrm{Var}(\hat{\theta})$, where we treat the data X as random. This will come into play later:

$$\begin{aligned}\mathrm{Var}(\hat{\theta}) &= \mathrm{Var}(\bar{X}_n)\\ &= \frac{1}{n^2}\mathrm{Var}\Big(\sum_{i=1}^{n} X_i\Big)\\ &= \frac{\sigma^2}{n}\\ &= \frac{1}{I(\hat{\theta})}.\end{aligned}$$

Later, we'll show that for any general MLE estimator:

$$\mathrm{Var}(\hat{\theta}) \approx \frac{1}{I(\hat{\theta})}.$$

This is a common theme: something that is exactly true in the normal case is approximately elsewhere. This is particularly true if the estimates are "regular", meaning they behave approximately like a normal distribution. □

- Calculating the observed Fisher's Information at any given point is rather simple: Find the likelihood, take second derivative(s), plug in desired point.
- However, because data are random, we may want to consider the *expected* Fisher's information.

Theorem: Expected Score

Now consider treating the score function $S(\theta)$ as a random function, dependent on random input data X. That is:

$$S(\theta) = \frac{\partial}{\partial\theta}\log f(X;\theta).$$

Then, under mild regularity conditions, the expected score is zero:

$$E\big[S(\theta)\big] = 0.$$

Proof.

$$E\Big[\frac{\partial}{\partial\theta}\log f(X;\theta)\Big] = \int_{\mathcal{X}} \frac{\partial}{\partial\theta}\log f(x;\theta) f(x;\theta)\,dx$$

$$\begin{aligned}
&= \int_{\mathcal{X}} \frac{f(x;\theta)}{f(x;\theta)}\frac{\partial}{\partial\theta} f(x;\theta)\,dx \\
&= \int_{\mathcal{X}} \frac{\partial}{\partial\theta} f(x;\theta) \\
&= \frac{\partial}{\partial\theta}\int_{\mathcal{X}} f(x;\theta)\,dx \\
&= \frac{\partial}{\partial\theta} 1 = 0.
\end{aligned}$$

Note that the conditions needed for this is the ability to swap the integral with the derivative, which is a condition that we briefly discussed last semester (Dominated Convergence Theorem, for instance). □

- *NOTE:* This is the most common way to define the score (and later, information). This holds for either a single observation, or the entire collection of data.

Definition: Fisher's Information

Continuing with the treatment of the score function as a random function of X, we will define the Expected Fisher's Information to be the following:

$$\mathcal{I}(\theta) = E\Big[\Big(\frac{\partial}{\partial\theta}\log f(X;\theta)\Big)^2\Big] = E\Big[\big(S(\theta)\big)^2\Big].$$

This is often referred to as just Fisher's Information.

Proposition: Variance of the Score

If the partial derivative and integral operations can be exchanged, then

$$\mathcal{I}(\theta) = \mathrm{Var}\big(S(\theta)\big) = -E\Big[\frac{\partial^2}{\partial\theta^2}\log f(X;\theta)\Big],$$

where the score function is treated as random.

Proof. If we can exchange integration and derivation, then the first equality in the statement is trivial. Under these conditions, we showed:

$$E\big[S(\theta)\big] = E\Big[\frac{\partial}{\partial\theta}\log f(X;\theta)\Big] = 0.$$

Therefore:

$$\begin{aligned}
\mathrm{Var}\big(S(\theta)\big) &= E\Big[S(\theta)^2\Big] - \Big(E\big[S(\theta)\big]\Big)^2 \\
&= \mathcal{I}(\theta).
\end{aligned}$$

Now if the second derivatives exist, the same argument that used to show the equality above can be used to show:

$$E\Big[\frac{\frac{\partial^2}{\partial\theta^2} f(X;\theta)}{f(X;\theta)}\Big] = \int_{\mathcal{X}} \frac{\partial^2}{\partial\theta^2} f(X;\theta)\,dx = 0.$$

(To do this, we exchange partial derivatives with the integral, note the integral is equal to one, which has partial derivative zero).

Building on these two results, note the following identity:

$$\begin{aligned}
\frac{\partial^2}{\partial\theta^2}\log f(X;\theta) &= \frac{\partial}{\partial\theta}\Big(\frac{\frac{\partial}{\partial\theta}f(X;\theta)}{f(X;\theta)}\Big) \\
&= \frac{f(X;\theta)\frac{\partial^2}{\partial\theta^2}f(X;\theta) - \big(\frac{\partial}{\partial\theta}f(X;\theta)\big)^2}{\big(f(X;\theta)\big)^2} \\
&= \frac{\frac{\partial^2}{\partial\theta^2}f(X;\theta)}{f(X;\theta)} - \Big(\frac{\frac{\partial}{\partial\theta}f(X;\theta)}{f(X;\theta)}\Big)^2 \\
&= \frac{\frac{\partial^2}{\partial\theta^2}f(X;\theta)}{f(X;\theta)} - \Big(\frac{\partial}{\partial\theta}\log f(X;\theta)\Big)^2
\end{aligned}$$

Taking expectation of both sides of this equation yields:

$$\begin{aligned}
E\Big[\frac{\partial^2}{\partial\theta^2}\log f(X;\theta)\Big] &= E\Big[\frac{\frac{\partial^2}{\partial\theta^2}f(X;\theta)}{f(X;\theta)}\Big] - E\Big[\Big(\frac{\partial}{\partial\theta}\log f(X;\theta)\Big)^2\Big] \\
&= -E\Big[\Big(\frac{\partial}{\partial\theta}\log f(X;\theta)\Big)^2\Big],
\end{aligned}$$

where the last inequality is a result of the previously derived identity

$$E\Big[\frac{\frac{\partial^2}{\partial\theta^2}f(X;\theta)}{f(X;\theta)}\Big] = 0.$$

Thus, we have proven the statement

$$\mathcal{I}(\theta) = \mathrm{Var}\big(S(\theta)\big) = -E\Big[\frac{\partial^2}{\partial\theta^2}\log f(X;\theta)\Big].$$

□

- This proposition is useful because:
  - If the log-likelihood is differentiable, the first definition $\mathcal{I}(\theta) = E[S(\theta)^2]$ always exists.
  - Typically, we can express this as $\mathcal{I}(\theta) = \mathrm{Var}(S(\theta))$, which is a nice interpretable quantity.
  - The final expression $\mathcal{I}(\theta) = -E\Big[\frac{\partial^2}{\partial\theta^2}\log f(X;\theta)\Big]$ is usually the easiest to calculate.
- For the multivariate case, this is equivalent to writing:

$$\begin{aligned}
\big[\mathcal{I}(\theta)\big]_{i,j} &:= E_\theta\Big[\frac{\partial\log f(X;\theta)}{\partial\theta_i}\frac{\partial\log f(X;\theta)}{\partial\theta_j}\Big] \\
&= \mathrm{Cov}\Big(\frac{\partial\log f(X;\theta)}{\partial\theta_i}, \frac{\partial\log f(X;\theta)}{\partial\theta_j}\Big) \\
&= -E\Big[\frac{\partial^2}{\partial\theta_i\partial\theta_j}\log f(X;\theta)\Big]
\end{aligned}$$

- Note that the curvature $I(\theta)$, defined as the second derivative of $L(\theta)$ (fixed function based on the observed data), exists for all points θ.

- Many theorem's instead rely on $\mathcal{I}(\theta)$, which is the *expected* Fisher's information. This later quantity "integrates out" the effect of the randomness in the data.
- In practice, the expected information is often hard to find, while the observed information is easy to calculate. For instance, in cases where the expected information is available, we might be able to get *exact* confidence intervals, which are preferred.
- In practice, we often replace $\mathcal{I}(\hat{\theta})$ in theory with $I(\hat{\theta})$.
- However, we're primarily only interested in this function at the MLE, *or* when we regard $S(\theta)$ as a random function of arbitrary X (not fixed on x^*).

Comment on numeric optimization
If numeric optimization is used to find the MLE, many approaches estimate the Hessian of the loss-function. If the negative log-likelihood is used, setting `optim(..., hessian = TRUE)` can be used to compute the Observed Fisher's Information (don't forget to adjust for variable transformations!)

3 Asymptotic properties of the MLE and Bayesian estimators

Normality of MLE

- Under weak regularity conditions, both the MLE and Bayesian posterior converge to a Normal distribution, centered at the true value.
- We'll begin with a sketch proof of the distribution of the MLE.

Theorem: Asymptotic distribution of the MLE
Let $x_1^*, x_2^*, \ldots, x_n^*$ come from a family of distributions with density $f(x;\theta)$. If the true value of θ is θ_0, i.e., $X_i \overset{iid}{\sim} f(x;\theta_0)$, and we denote $\hat{\theta}$ as the MLE of θ, then

$$\sqrt{n\mathcal{I}(\theta_0)}\big(\hat{\theta} - \theta_0\big) \xrightarrow{d} Z$$

where $Z \sim N(0,1)$.

Proof. Recall in this setup, we have X being multivariate. Thus, the likelihood function is

$$L(\theta) := \prod_{i=1}^{n} f(X_i;\theta).$$

Note that here, we are considering the long-run distribution of the MLE estimator, so X_i is treated as random, not fixed.
The theorem statement depends on $\mathcal{I}(\theta_0)$, which is defined as the (expectation of the) negative second derivative of the log-likelihood, evaluated at θ_0.

$$\ell(\theta) = \sum_{i=1}^{n} \log f(X_i;\theta).$$

By definition,

$$\hat{\theta} = \operatorname*{argmax}_{\theta} \ell(\theta),$$

for any fixed $X = x^*$. Thus, if the MLE is not a boundary point, and the likelihood function is differentiable, then

$$\ell'(\hat{\theta}) = 0.$$

Now consider a Taylor-series expansion of $\ell'(\hat{\theta})$ around the point θ_0:

$$0 = \ell'(\hat{\theta}) = \ell'(\theta_0) + \big(\hat{\theta} - \theta_0\big)\ell''(\theta_0) + \frac{1}{2}\big(\hat{\theta} - \theta_0\big)^2 \ell'''(\theta^*),$$

where θ^* is some point between $\hat{\theta}$ and θ_0.
Now we want to rearrange so that on the left hand side, we get the expression that we are looking for: $\sqrt{nI(\theta_0)}\big(\hat{\theta} - \theta_0\big)$. Subtracting away terms involving $\big(\hat{\theta} - \theta_0\big)$,

$$-\big(\hat{\theta} - \theta_0\big)\Big(\ell''(\theta_0) + \frac{1}{2}\big(\hat{\theta} - \theta_0\big)\ell'''(\theta^*)\Big) = \ell'(\theta_0),$$

and multiplying by $\sqrt{n}$, we have:

$$\begin{aligned}\sqrt{n}\big(\hat{\theta} - \theta_0\big) &= \frac{\sqrt{n}\ell'(\theta_0)}{-\ell''(\theta_0) - \frac{1}{2}\big(\hat{\theta} - \theta_0\big)\ell'''(\theta^*)} \\ &= \frac{\frac{1}{\sqrt{n}}\ell'(\theta_0)}{-\frac{1}{n}\ell''(\theta_0) - \frac{1}{2n}\big(\hat{\theta} - \theta_0\big)\ell'''(\theta^*)}.\end{aligned}$$

The last step seems a bit odd, but it will be used to apply Slutsky's Theorem (covered last semester) to finish the proof. As a reminder, if W_n is a sequence of random variables that converges in *distribution* to W, and Y_n is a sequence of random variables that convergence in *probability* to a constant c, then $W_n/Y_n \xrightarrow{d} W/c$. Thus, we want something like:

$$\sqrt{n}\big(\hat{\theta} - \theta_0\big) = W_n/Y_n,$$

where W_n converges in distribution to some normally distributed random variable, and Y_n converges to a constant.
We have expressed the numerator such that:

$$\frac{1}{\sqrt{n}}\ell'(\theta_0) = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{\partial}{\partial\theta}\log f(X_i;\theta_0).$$

Since $E[\ell'(\theta_0)] = E[S(\theta)] = 0$, the central limit theorem states:

$$\frac{1}{\sqrt{n}}\ell'(\theta_0) \xrightarrow{d} N(0, \mathcal{I}(\theta_0)).$$

In the denominator, the leading term converges in probability to $I(\theta_0)$ by the weak law of large numbers:

$$-\frac{1}{n}\ell''(\theta_0) \xrightarrow{p} E\Big[-\frac{\partial^2}{\partial\theta^2}\log f(X_i;\theta_0)\Big] = \mathcal{I}(\theta_0).$$

Now we need to be a bit careful with the remainder term. The argument follows closely that which was used to prove the CLT last semester. This does involve a few "regularity" conditions, under which we can show that the remainder term converges to zero in probability. In this case, the entire denominator converges to zero in probability. The additional conditions needed are:

- bounded third derivative: $|\ell'''(\theta)| \leq M(X)$ for all θ, where $E[M(X)] < \infty$.

This last identity is needed because the remainder is:

$$R_n = \frac{1}{2n}\big(\hat{\theta} - \theta_0\big)\sum_{i=1}^{n}\ell'''(X_i;\theta^*).$$

Later in the semester we will formally show that $\big(\hat{\theta} - \theta_0\big) \xrightarrow{p} 0$. If the third derivative is bounded by $M(X)$, which has finite expectation, the strong law of large numbers implies that $\frac{1}{2n}\sum_i \ell'''(X_i;\theta^*)$ is bounded, and because $\big(\hat{\theta} - \theta_0\big) \xrightarrow{p} 0$, the entire remainder term $R_n \xrightarrow{p} 0$.
Finally, this combined work shows that the denominator in the expression for $\sqrt{n}\big(\hat{\theta} - \theta_0\big)$ is:

$$Y_n = -\frac{1}{n}\ell''(\theta_0) - \frac{1}{2n}\big(\hat{\theta} - \theta_0\big)\ell'''(\theta^*) \xrightarrow{p} \mathcal{I}(\theta_0),$$

and therefore:

$$\sqrt{n}\big(\hat{\theta} - \theta_0\big) = W_n/Y_n \xrightarrow{p} W/\mathcal{I}(\theta_0) \stackrel{d}{=} N(0, 1/\sqrt{\mathcal{I}(\theta_0)}),$$

or by multiplying by $\sqrt{\mathcal{I}(\theta_0)}$, we get the theorem statement:

$$\sqrt{n\mathcal{I}(\theta_0)}\big(\hat{\theta} - \theta_0\big) \xrightarrow{d} Z,$$

where $Z \sim N(0, 1)$. □

- How do we interpret / use the result above?
- *For large iid samples:* the MLE $\hat{\theta}$ is approximately normally distributed, centered at the *true* value θ_0, with shrinking variance:
$$\mathrm{Var}(\hat{\theta}) = \frac{1}{n\mathcal{I}(\theta_0)}.$$
- *Note* that there is sometimes some confusion regarding whether or not $\mathcal{I}$ is the information from *a single* observation, or *all observations*.
- Because the data are IID, it's really just a difference in scaling constants.
- In our definition and proof, we let $\mathcal{I}(\theta_0)$ be the information from a single observation:
$$\mathcal{I}(\theta_0) = E\Big[\Big(\frac{\partial}{\partial\theta}\log f(X_i;\theta)\Big)^2\Big].$$
- We could have instead considered the likelihood / information using all of the data points:
$$\mathcal{I}_n(\theta_0) = E\Big[\Big(\frac{\partial}{\partial\theta}\log f_{X_1:X_n}(X_1,\ldots,X_n;\theta)\Big)^2\Big].$$
- In the later case, the theorem would have stated:
$$\sqrt{\mathcal{I}_n}\big(\hat{\theta} - \theta_0\big) \xrightarrow{d} Z \sim N(0, 1),$$
or that
$$\mathrm{Var}(\hat{\theta}) = \frac{1}{\mathcal{I}_n(\theta_0)}$$
- In the IID case, these are the same, as
$$\begin{aligned}\mathcal{I}_n(\theta) &= -E\Big[\frac{\partial^2}{\partial\theta^2}\sum_i \log f(X_i;\theta)\Big] \\ &= -\sum_i E\Big[\frac{\partial^2}{\partial\theta^2}\log f(X_i;\theta)\Big] = n\mathcal{I}(\theta).\end{aligned}$$
- Our derivation treated θ as univariate. The more common multivariate case has a similar result.

Multivariate parameters
If $\theta \in \mathbb{R}^d$ is multivariate, than using the matrix definition of $\mathcal{I}(\theta)$, then:

$$\sqrt{n}\big(\hat{\theta} - \theta_0\big) \xrightarrow{d} Z_d \sim N_d\big(0_d, \mathcal{I}(\theta_0)^{-1}\big),$$

where $\hat{\theta}$ is the MLE, θ_0 is the true parameter value, and $N_d(\mu, \Sigma)$ is a d-dimensional multivariate normal distribution with mean vector μ and covariance matrix Σ.

Implications for estimating parameter uncertainty

- This result gives us an immediate way to approximate the uncertainty associated with the MLE, for rather arbitrary models.
- If $\hat{\theta}$ is the MLE, then:
$$\mathrm{Var}\big(\hat{\theta}\big) \approx \frac{1}{n\mathcal{I}(\hat{\theta})} = \frac{1}{\mathcal{I}_n(\hat{\theta})}.$$
- *There are two different approximations here*:
  - Asymptotic normality.
  - $\hat{\theta} \xrightarrow{p} \theta_0$ (Slutksy's Theorem).
- It is a good approximation for even moderate n, but exact calculations are preferred when possible.
- Finally, recall that $\mathcal{I}(\hat{\theta})$ is an *expectation*.
- When an exact calculation is not readily available, calculating the expectation $\mathcal{I}(\hat{\theta})$ usually isn't either!
- In practice, we often replace the Fisher's Information with the Observed Fisher's information:
$$\mathrm{Var}\big(\hat{\theta}\big) \approx \frac{1}{I_n(\hat{\theta})},$$
where I_n is the information provided by all n data points.
- The details justifying this final approximation are a bit tricky. See Keener (2010, Chapter 9) for more details.
- In short, for many distributions (regular exponential families), $\mathcal{I}(\hat{\theta}) = I(\hat{\theta})$. In other cases, suitable conditions imply that $I(\hat{\theta}) \xrightarrow{p} \mathcal{I}(\hat{\theta})$.

Asymptotic results for Bayesian posteriors

- Similar results hold for Bayesian estimates.

Bernstein–von Mises Theorem
We are not ready to state the full Theorem, so we'll just provide a heuristic approach: under suitable conditions, the posterior distribution behaves asymptotically like a normal distribution with mean $\hat{\theta}_{MLE}$, and variance $1/\mathcal{I}(\theta_0)$.

Proof. (Sketch, not a formal proof).
Let the collection of observations be denoted X, and the parameter of interest θ. Then:

$$\begin{aligned}\pi_{\Theta|X}(\theta|x) &\propto \pi_\Theta(\theta) f_{X|\Theta}(x|\theta) \\ &= \exp\big[\log \pi_\Theta(\theta)\big] \exp\big[\log f_{X|\Theta}(x|\theta)\big] \\ &= \exp\big[\log \pi_\Theta(\theta)\big] \exp\big[\ell(\theta)\big]\end{aligned}$$

We will now again perform a Taylor-Series expansion of the log-likelihood, about the MLE $\hat{\theta}$. With enough observations, this expansion will be very accurate. In the case where we have large ammounts of data, the likelihood "dominates" over the prior, negating the effect of the prior distribution. Thus:

$$\begin{aligned}\pi_{\Theta|X}(\theta|x) &\propto \exp\big[\log \pi_\Theta(\theta)\big] \exp\big[\ell(\theta)\big] \\ &\approx \exp\big[\ell(\theta)\big] \\ &\approx \exp\Big[\ell(\hat{\theta}) + (\theta - \hat{\theta})\ell'(\hat{\theta}) + \frac{1}{2}(\theta - \hat{\theta})^2 \ell''(\hat{\theta})\Big] \\ &\propto \exp\Big[\frac{1}{2}(\theta - \hat{\theta})^2 \ell''(\hat{\theta})\Big].\end{aligned}$$

In this last expression, we get what looks like the density of a normal distribution, centered at $\hat{\theta}$, with variance $-[\ell''(\hat{\theta})]^{-1} = 1/I(\hat{\theta})$. □

4 Bias, MSE, and Consistency

Bias

- We have already talked a little about bias last semester.
- It's an important concept / principle for point-estimates.

  Definition: Bias of an estimator

  The *bias* of a point estimator $\hat{\theta}$ of a parameter θ is the difference between the expected value of $\hat{\theta}$ and θ; that is,

  $$\text{Bias}_\theta\big[\hat{\theta}\big] = E_\theta\big[\hat{\theta}\big] - \theta$$

- If $\text{Bias}_\theta\big[\hat{\theta}\big] = 0$, then $\hat{\theta}$ is an *unbiased* estimator of θ, and $E_\theta\big[\hat{\theta}\big] = 0$.
- *unbiasedness* is a nice property, but alone does not mean we have a good estimator.
- Another thing that needs to be considered is the *variance* of an estimator.
- We encountered this with Monte-Carlo methods: many approaches converge to the true value, but lower variance estimates are generally preferred among unbiased methods.

Unbiased estimators for normal distribution
Let $X_1, X_2, \ldots, X_n$ be iid from $N(\mu, \sigma^2)$. Consider two estimates of the parameter $\theta = \mu$:

$$\hat{\theta}_1 = \bar{X}_n, \qquad \hat{\theta}_2 = X_1.$$

Show that both are unbiased estimators, but that $\hat{\theta}_1$ has lower variance.

MSE

- The mean square error (MSE) describes the expected squared difference between an estimator and the true parameter value.
- Often, attempts as lowering the *variance* results in increasing the *variance*.
- It can be used to balance the bias-variance tradeoff.

Definition + Theorem: MSE
Let $\hat{\theta}$ be an estimator of θ. The MSE of $\hat{\theta}$ is defined as:

$$\mathrm{MSE}_\theta\big(\hat{\theta}\big) := E\big[(\hat{\theta}-\theta)^2\big].$$

Furthermore, $\mathrm{MSE}_\theta\big(\hat{\theta}\big) = \mathrm{Var}_\theta\big(\hat{\theta}\big) + \big(\mathrm{Bias}_\theta(\hat{\theta})\big)^2$.

Proof.

$$\begin{aligned}
\mathrm{MSE}_\theta\big(\hat{\theta}\big) &= E\big[(\hat{\theta}-\theta)^2\big] \\
&= E\big[(\hat{\theta}-E[\hat{\theta}]+E[\hat{\theta}]-\theta)^2\big] \\
&= E\big[(\hat{\theta}-E[\hat{\theta}])^2 + 2(\hat{\theta}-E[\hat{\theta}])(E[\hat{\theta}]-\theta) + (E[\hat{\theta}]-\theta)^2\big] \\
&= E\big[(\hat{\theta}-E[\hat{\theta}])^2\big] + 2E\big[(\hat{\theta}-E[\hat{\theta}])(E[\hat{\theta}]-\theta)\big] + E\big[(E[\hat{\theta}]-\theta)^2\big] \\
&= E\big[(\hat{\theta}-E[\hat{\theta}])^2\big] + 2E\big[(\hat{\theta}-E[\hat{\theta}])\big](E[\hat{\theta}]-\theta) + (E[\hat{\theta}]-\theta)^2 \\
&= E\big[(\hat{\theta}-E[\hat{\theta}])^2\big] + 2(E[\hat{\theta}]-E[\hat{\theta}])(E[\hat{\theta}]-\theta) + (E[\hat{\theta}]-\theta)^2 \\
&= E\big[(\hat{\theta}-E[\hat{\theta}])^2\big] + (E[\hat{\theta}]-\theta)^2 \\
&= \mathrm{Var}_\theta\big(\hat{\theta}\big) + \big(\mathrm{Bias}_\theta(\hat{\theta})\big)^2
\end{aligned}$$

□

Example: Variance of MLE
Let X_i be iid from $N(\mu,\sigma^2)$, and consider estimating $\theta = (\mu,\sigma^2)$ in two ways:

$$\hat{\theta}_{MLE} = \Big(\bar{X}_n, \hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^n \big(\bar{X}-\sigma^2\big)^2\Big),$$

$$\hat{\theta}_1 = \Big(\bar{X}_n, S^2 = \frac{1}{n-1}\sum_{i=1}^n \big(\bar{X}-\sigma^2\big)^2\Big).$$

What is the MSE of $\hat{\mu}_{MLE}$? What about the bias and MSE of S^2 and $\hat{\sigma}^2$?
We will use the Bias-Variance characterization of MSE. First, consider the estimate $\hat{\mu} = \bar{X}_n$, which is the same for both of the proposed methods. First, we will show it is an unbiased estimator for μ, and then find the variance of this estimator.

$$\begin{aligned}
E\big[\bar{X}_n\big] &= E\Big[\frac{1}{n}\sum_i X_i\Big] \\
&= \frac{1}{n}\sum_i E[X_i] \\
&= \frac{n}{\mu n} = \mu.
\end{aligned}$$

Thus, $E\big[\hat{\mu} - \mu\big] = 0$, and $\hat{\mu}$ is an unbiased estimator. Thus,

$$\begin{aligned}
\mathrm{MSE}_\mu\big(\hat{\mu}\big) &= \mathrm{Var}\big(\hat{\mu}\big) - \Big(\mathrm{Bias}_\mu\big(\hat{\mu}\big)\Big)^2 \\
&= \mathrm{Var}\big(\hat{\mu}\big) \\
&= \mathrm{Var}\Big(\frac{1}{n}\sum_i X_i\Big) \\
&= \frac{1}{n^2}\sum_i \mathrm{Var}(X_i) \\
&= \frac{\sigma^2}{n}
\end{aligned}$$

Now consider the two different estimates for σ^2, either the MLE $\hat{\sigma}^2_{MLE}$, and the sample variance S^2. From Homework 7, Problem 5 of last semester, you showed:

$$\begin{aligned}
E\big[\hat{\sigma^2}_{MLE}\big] &= E\Big[\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2\Big] \\
&= \frac{n-1}{n}\sigma^2,
\end{aligned}$$

and, by extension, you derived the estimator:

$$E\big[S^2\big] = E\Big[\frac{n}{n-1}\hat{\sigma}^2_{MLE}\Big] = \frac{n}{n-1}\frac{n-1}{n}\sigma^2 = \sigma^2$$

Thus, we see that $\hat{\sigma}^2_{MLE}$ is a *biased* estimator, while S^2 is unbiased. The bias of the MLE can be found as:

$$E\big[\hat{\sigma^2}_{MLE} - \sigma^2\big] = \Big(\frac{n-1}{n}\sigma^2 - \sigma^2\Big).$$

From Chapter 6 last semester, we proved that if X_i are iid normal, then $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$. This was used to show that $(\bar{X} - \mu)/(S/\sqrt{n})$ has a t-distribution. Thus:

$$\mathrm{Var}\big((n-1)S^2/\sigma^2\big) = \mathrm{Var}(Z) = 2(n-1),$$

where $Z = (n-1)S^2/\sigma^2$ follows a chi-square distribution. Therefore

$$\mathrm{Var}\big(S^2\big) = \mathrm{Var}\Big(\frac{\sigma^2}{n-1}Z\Big) = \frac{\sigma^4}{(n-1)^2}2(n-1) = \frac{2\sigma^4}{n-1}.$$

Since S^2 is an unbiased estimator of σ^2, we have:

$$\mathrm{MSE}_{\sigma^2}\big(S^2\big) = \mathrm{Var}\big(S^2\big) = \frac{2\sigma^4}{n-1}.$$

We can extend this result to find the variance and MSE of $\hat{\sigma}^2_{MLE}$. First, the variance:

$$\mathrm{Var}\big(\hat{\sigma}^2_{MLE}\big) = \mathrm{Var}\Big(\frac{n-1}{n}S^2\Big) = \Big(\frac{n-1}{n}\Big)^2\mathrm{Var}\big(S^2\big) = \frac{2(n-1)\sigma^4}{n^2}.$$

The MSE is then given by:

$$\begin{aligned}
\mathrm{MSE}_{\sigma^2}\big(\hat{\sigma}^2_{MLE}\big) &= \frac{2(n-1)\sigma^4}{n^2} + \Big(\frac{n-1}{n}\sigma^2 - \sigma^2\Big)^2 \\
&= \Big(\frac{2n-1}{n^2}\Big)\sigma^2.
\end{aligned}$$

Comparing this MSE to the MSE of S^2, it can be shown that for all n,

$$\mathrm{MSE}_{\sigma^2}\big(\hat{\sigma}^2_{MLE}\big) < \mathrm{MSE}_{\sigma^2}\big(S^2\big),$$

or that although the MLE is biased for σ^2, it has a lower MSE than S^2.

- Based on the examples above, should $\hat{\sigma^2}$ be used instead of S^2? The argument above proves that, on average, $\hat{\sigma}^2_{MLE}$ will be closer to σ^2 than S^2.
- *However*: statisticians generally agree that S^2 is a better estimate. Why?
- Though it has lower MSE, *on average*, $\hat{\sigma}^2_{MLE}$ will be *smaller* than σ^2 (biased low).
- There's also some agreement that MSE is *not* a reasonable criterion for *scale* parameters, like the variance.
- The reason is that the MSE is a symmetric criterion: both small and large errors are penalized the same.
- However, scale parameters like σ^2 must be positive, so they have a natural lower bound $\sigma^2 > 0$, so the estimation problem is *not* symmetric.
- The MSE_θ uses the subscript θ to denote that it is a function of θ.
- Often, there is not a *uniformly best* estimator for θ, and some estimators will perform better for some values of θ and worse for others.

MSE of MLE and Bayes: Binomial Model
(Example 7.3.5 Casella and Berger, 2024) Let X_i be iid Bernoulli(p), such that $X = \sum_i X_i$ is Binomial(n, p). The MLE $\hat{p} = X/n$ is unbiased for p, and has MSE:

$$\mathrm{MSE}_p\big(\hat{p}\big) = \mathrm{Var}_p\big(\hat{p}\big) = \frac{p(1-p)}{n}.$$

Compare this to the posterior mean estimate, where the prior $P \sim U(0,1)$.
The posterior distribution is:

$$P|X \sim \mathrm{Beta}\big(X+1, n-X+1\big),$$

which has posterior mean:

$$\hat{p}_B = E[P|X] = \frac{X+1}{n+2},$$

The MSE of this estimate is:

$$\begin{aligned}\mathrm{MSE}_p\big(\hat{p}_B\big) &= \mathrm{Var}_p\big(\hat{p}_B\big) + \Big(\mathrm{Bias}_p\big(\hat{p}_B\big)\Big)^2 \\ &= \frac{np(1-p)}{(n+2)^2} + \Big(\frac{np+1}{n+2} - p\Big)^2\end{aligned}$$

We may want to consider plotting the MSE, as a function of p, for fixed values of n. Below are two plots for $n = 4$ and $n = 400$:

- https://www.desmos.com/calculator/j2hjnqnqpc
- https://www.desmos.com/calculator/skeej4pndh

Interestingly, at $n = 4$, the MSE of $\hat{p}_B$ is constant. In both figures, we see that if the true value of p is close to 0.5, then the MSE of $\hat{p}_B$ is lower. However, for values of p close to 0 or 1, the MSE of $\hat{p}_{MLE}$ is lower. Neither estimate is *uniformly* better in terms of MSE. Note that for all values of p and n, $\hat{p}_B$ is biased.

Uniformly Best Unbiased Estimators

- As the previous example demonstrates, there is almost *never* a uniformly best estimate in terms of MSE.
- This is because the class of estimators is "too large".
- Example: consider estimating θ, using $\hat{\theta} = 17$. If $\theta = 17$, this will always beat any other estimator. However, it's a very poor estimate if $\theta \neq 17$.
- This leads us to thinking of ways of limiting the possible estimators to certain classes of estimators.
- A natural constraint to consider is the class of *unbiased estimators*. In this class, the MSE = Variance of the estimator.

Definition: Uniformly best unbiased estimators
An estimator θ^* is called the *best unbiased estimator* of θ if $E[\theta^*] = \theta$, for any other estimator $\tilde{\theta}$ such that $E[\tilde{\theta}] = \theta$, then $\mathrm{Var}_\theta\big(\theta^*\big) \leq \mathrm{Var}_\theta\big(\tilde{\theta}\big)$ for all θ.
θ^* is also called a *uniform minimum variance unbiased estimator* (UMVUE) of θ.

- Initially, it can be extremely difficult to find a UMVUE estimator without adidtional supporting theory.

Poisson unbiased estimation
Let X_i be iid Poisson(λ) random variables, and let $\bar{X}$ and S^2 be the sample mean and variance, respectively. Since $E[X_i] = \mathrm{Var}(X_i) = \lambda$, it can be shown that $E[\bar{X}] = E[S^2] = \lambda$, and therefore both $\bar{X}$ and S^2 are unbiased estimators of λ. Which is a better estimate?

- Since they are unbiased, we just need to find the lower variance of the two. Evaluating $\mathrm{Var}_\lambda\big(S^2\big)$ is actually a very lengthy calculation.
- However, even if we do the calculation and find $\mathrm{Var}_\lambda(\bar{X}) \leq \mathrm{Var}_\lambda(S^2)$, this doesn't immediately imply that $\bar{X}$ is the UMVUE. Consider for instance:

$$\hat{\lambda} = a\bar{X} + (1-a)S^2.$$

- This estimate is also unbiased, so from this alone we have an infinite number of unbiased estimators to consider.
- The next Theorem provides a *lower bound* on the variance of an unbiased estimator.

Theorem: Cramér-Rao Inequality
Let $X_1, \ldots, X_n$ be a sample with joint pdf $f_\theta(X;\theta)$, and let θ^* be any estimator of θ (biased or unbiased), that satisfies the smoothness condition

$$\frac{d}{d\theta} E_\theta\big[\theta^*\big] = \int_{\mathcal{X}} \frac{\partial}{\partial\theta} \theta^* f_\theta(X;\theta)\, dx.$$

Then,

$$\mathrm{Var}_\theta\big(\theta^*\big) \geq \frac{\big(\frac{d}{d\theta} E_\theta[\theta^*]\big)^2}{E_\theta\Big[\big(\frac{\partial}{\partial\theta} \log f(x;\theta)\big)^2\Big]} = \frac{\big(\frac{d}{d\theta} E_\theta[\theta^*]\big)^2}{\mathcal{I}(\theta)}.$$

Proof. The proof relies on the Cauchy-Schwarz inequality for random variables X and Y:

$$\big(\mathrm{Cov}(Z,Y)\big)^2 \leq \big(\mathrm{Var}(Z)\big)\big(\mathrm{Var}(Y)\big),$$

which implies:

$$\mathrm{Var}(Z) \geq \frac{\big(\mathrm{Cov}(Z,Y)\big)^2}{\mathrm{Var}(Y)}.$$

Then, we'll substitute $Z = \theta^*$ and $Y = \frac{\partial}{\partial\theta}\log f(X;\theta)$. First, using the condition in the Theorem that we can exchange the derivative with the integral:

$$\begin{aligned}
\frac{d}{d\theta}E_\theta\big[\theta^*\big] &= \int_{\mathcal{X}} \theta^* \frac{\partial}{\partial\theta} f(x;\theta)\,dx \\
&= \int_{\mathcal{X}} \theta^* \frac{\frac{\partial}{\partial\theta} f(x;\theta)}{f(x;\theta)} f(x;\theta)\,dx \\
&= \int_{\mathcal{X}} \theta^* \frac{\partial}{\partial\theta}\log f(x;\theta) f(x;\theta)\,dx \\
&= E_\theta\Big[\theta^* \frac{\partial}{\partial\theta}\log f(X;\theta)\Big].
\end{aligned}$$

Now since $E\Big[\frac{\partial}{\partial\theta}\log f(X;\theta)\Big] = 0$ (see theorem about expected score), we now make the substitution $Z = \theta^*$ and $Y = \frac{\partial}{\partial\theta}\log f(X;\theta)$.
Under these assumptions, the expected score is $E[Y] = 0$, and therefore $\mathrm{Cov}(Z,Y) = E[XY]$. Similarly, these same conditions imply $\mathrm{Var}(Y) = E_\theta\Big[\big(\frac{\partial}{\partial\theta}\log f(x;\theta)\big)^2\Big] = \mathcal{I}(\theta)$.
Thus, using the Cauchy-Schwarz inequality, and noticing:

$$\frac{d}{d\theta}E_\theta\big[\theta^*\big] = \mathrm{Cov}(Z,Y)$$

$$\mathrm{Var}(Z) = \mathrm{Var}(\theta^*) \geq \frac{\mathrm{Cov}(Z,Y)}{\mathrm{Var}(Y)} = \frac{\big(\frac{d}{d\theta}E_\theta[\theta^*]\big)^2}{E_\theta\Big[\big(\frac{\partial}{\partial\theta}\log f(x;\theta)\big)^2\Big]} = \frac{\big(\frac{d}{d\theta}E_\theta[\theta^*]\big)^2}{\mathcal{I}(\theta)}$$

□

- In the Cramér-Rao inequality, consider what happens if θ^* is an unbiased estimator of θ:

$$\Big(\frac{d}{d\theta}E_\theta[\theta^*]\Big)^2 = \Big(\frac{d}{d\theta}\theta\Big)^2 = 1.$$

- This leads to the common "Cramér-Rao lower bound", for unbiased estimators:

$$\mathrm{MSE}_\theta\big(\theta^*\big) = \mathrm{Var}_\theta\big(\theta^*\big) \geq \frac{1}{\mathcal{I}(\theta)}.$$

- The Cramér-Rao Inequality can at times be used to find UMVUEs (which are not necessarily unique).

Example: UMVUE for Poisson model
Let X_i be iid from a Poisson(λ) distribution. Both $\bar{X}$ and S^2 (the sample mean and variance, respectively) are unbiased estimators of λ. This implies that any estimator $a\bar{X} + (1-a)S^2$ with $a \in [0,1]$ is an unbiased estimator, among other possibilities.

If θ^* is *some* unbiased estimator, find the minimal possible variance of θ^*, and use that to find a UMVUE estimator of θ.
First we need to find the expected Fisher's information. Since the data are iid, we have $\mathcal{I}_n(\theta)$, denoting the information for the entire collection of observations $X_1, X_2, \ldots, X_n$, is equal to:

$$\mathcal{I}_n(\theta) = n\mathcal{I}(\theta)$$

where $\mathcal{I}(\theta)$ is the information for a single observation. Using the second characterization of Fisher's Information (i.e., using the fact that we can exchange derivative and integral for this family):

$$\begin{aligned}
\mathcal{I}(\theta) &= -E\Big[\frac{\partial^2}{\partial\theta^2}\log f(X_i;\theta)\Big] \\
&= -E\Big[\frac{\partial^2}{\partial\theta^2}\log\frac{e^{-\lambda}\lambda_i^X}{X_i!}\Big] \\
&= -E\Big[\frac{\partial^2}{\partial\theta^2}\big(-\lambda + X_i\log\lambda - \log X_i!\big)\Big] \\
&= -E\Big[\frac{\partial^2}{\partial\theta^2}\big(-1 + X_i/\lambda\big)\Big] \\
&= -E\Big[-\frac{X_i}{\lambda^2}\Big] \\
&= \frac{1}{\lambda^2}E[X_i] \\
&= \frac{1}{\lambda}.
\end{aligned}$$

Thus, for all of the observations, the Fisher's information is given by:

$$\mathcal{I}_n(\theta) = \frac{n}{\lambda}.$$

Therefore, for every possible unbiased estimator λ^*, the Cramér-Rao inequality states that:

$$\mathrm{Var}(\lambda^*) \geq \frac{\lambda}{n}.$$

Notice that $\hat{\lambda} = \bar{X}_n$, the MLE of λ, is unbiased:

$$E\big[\hat{\lambda}\big] = E\big[\bar{X}_n\big] = \lambda,$$

and

$$\mathrm{Var}\big(\hat{\lambda}\big) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{\lambda}{n},$$

which is equal to the Cramér-Rao lower bound! This implies that for *any* unbiased estimator λ^*, we have:

$$\mathrm{Var}(\lambda^*) \geq \mathrm{Var}(\hat{\lambda}).$$

Therefore, for the iid Poisson(λ) model, the MLE $\hat{\lambda} = \bar{X}_n$ is a UMVUE of λ.

- One weakness of the Cramér-Rao inequality is that the lower bound, $\frac{1}{\mathcal{I}(\theta)}$, *might not be tight.*
- That is, it gives us a minimum quantity to compare to, but there may not exist an unbiased estimator that *attains* the lower-bound.
- There are attainment theorems, stating when the Cramér-Rao inequality is attainable and when it is not. We won't present any of these in this class, but it's closely related to *exponential families.*
- Instead, we can use some of our existing theorems to provide asymptotic results.

Consistency

- Before continuing our discussion about asymptotic results, we will introduce a few new definitions and principles of point estimation.

Definition: Consistency
Estimators of θ are really sequences of random variables: θ_n^*. Example: estimate $E[X_i] = \mu$ with $\bar{X}_n = \frac{1}{n}\sum_i X_i$.
One desirable feature of a sequence of estimators θ_n^* is that as n grows large, θ_n^* converges in some way to the truth θ.
We say that an estimator θ_n^* is *consistent* if it converges in probability to the true value θ:

$$\lim_{n\to\infty} P_\theta\big(|\theta_n^* - \theta| \geq \epsilon\big) = 0.$$

- When trying to show convergence in probability last semester, we often used Chebychev's inequality:

$$P_\theta\big(|\theta_n^* - \theta| \geq \epsilon\big) \leq \frac{E_\theta\Big[(\theta_n^* - \theta)^2\Big]}{\epsilon^2}.$$

- Thus, if for every value of θ, we have:

$$\lim_{n\to\infty} E_\theta\Big[(\theta_n^* - \theta)^2\Big] = 0,$$

  then we know that θ_n^* is a consistent estimator of θ.

- Note that this condition is our definition of MSE!

$$\mathrm{MSE}_\theta\big(\theta_n^*\big) = E_\theta\Big[(\theta_n^* - \theta)^2\Big] = \mathrm{Var}_\theta\big(\theta_n^*\big) + \Big(\mathrm{Bias}_\theta\big(\theta_n^*\big)\Big)^2.$$

- From this, we get an immediate result:

Theorem: Convergence in MSE
If θ_n^* is a sequence of estimators for θ, then if for every value of θ,

1. $\lim_{n\to\infty} \mathrm{Var}_\theta\big(\theta_n^*\big) = 0$
2. $\lim_{n\to\infty} \mathrm{Bias}_\theta\big(\theta_n^*\big) = 0$

then θ_n^* is a consistent estimator for θ.

Consistency of the MLE
Under suitable conditions on the likelihood function, the MLE $\hat{\theta}$ is a consistent estimator of θ.
A simple application of the Bernstein-von Misses Theorem implies that the Bayesian posterior mean is also a consistent estimator of θ.
(Proof sketch)
We previously showed that, asymptotically, the MLE satisfies:

$$\hat{\theta} \approx N\Big(\theta, \frac{1}{n\mathcal{I}(\theta)}\Big),$$

where $\mathcal{I}(\theta)$ is the Fisher's information for a single observation. This is a fixed quantity, and thus we see that the limiting expectation $\lim_{n\to\infty} E[\hat{\theta}] = \theta$, so the $\mathrm{Bias}_\theta(\hat{\theta}) \to 0$. Similarly,

$$\mathrm{Var}(\hat{\theta}) \approx \frac{1}{nI(\theta)} \to 0.$$

Efficiency

Definition: relative efficiency
If θ^* and $\tilde{\theta}$ are two estimators of θ, then the *relative efficiency* of θ^* with respect to $\tilde{\theta}$ is defined as:

$$\mathrm{RE}\big(\theta^*, \tilde{\theta}\big) = \frac{\mathrm{Var}\big(\tilde{\theta}\big)}{\mathrm{Var}\big(\theta^*\big)}.$$

- If RE > 1, then θ^* is said to be a more efficient estimator than $\tilde{\theta}$
- If RE < 1, then θ^* is said to be less efficient than $\tilde{\theta}$.
- If θ^* is more efficient than any other estimator, we call θ^* an *efficient estimator*.

Definition: Limiting variance and relative efficiency
An estimator θ_n^* of θ depends on the data, and the number of observations. If $\lim_{n\to\infty} k_n\theta^* = \sigma^2$ for some sequence k_n and constant σ^2, then σ^2 is called the *limiting variance of the estimator* θ^*.
If $\tilde{\theta}_n^*$ is a different estimate of θ with limiting variance τ^2, then the *relative asymptotic efficiency* of θ_n^* with respect to $\tilde{\theta_n}$ is defined as:

$$\mathrm{ARE}\big(\theta_n^*, \tilde{\theta}\big) = \frac{\tau^2}{\sigma^2}$$

Theorem: asymptotic efficiency of MLE and posterior mean
Under suitable regularity conditions, MLE $\hat{\theta}$ for parameter vector θ is *asymptotically efficient*, achieving the Cramér-Rao lower bound $\mathcal{I}(\theta)$.
By the Bernstein-von Misses Theorem, posterior distribution in a Bayesian analysis asymptotically converges to a normal distribution with mean $\hat{\theta}$, and variance $1/\mathcal{I}(\theta)$. Consequently, the posterior mean is also asymptotically efficient.

Proof. This Theorem is an immediate consequence of the asymptotic normality characterization of the MLE. Specifically, if the likelihood is twice differentiable and we can exchange the order of integration and derivative, then

$$\sqrt{n}\big(\hat{\theta}_{MLE} - \theta\big) \xrightarrow{d} Z \sim N\Big(0, \frac{1}{\mathcal{I}(\theta)}\Big).$$

The statement regarding the Bayesian posterior mean is similarly trivial given the Bernstein-von Misses Theorem. □

- *Brief Summary*:
- We have argued that under very mild conditions, the MLE and Bayesian posterior mean are both *consistent* (i.e., converges in probability to the true parameter value), **and** *asymptotically efficient*.
- This largely explains why these two approaches dominate the literature on parameter estimation:
- even if closed-form expressions for the sampling distributions of parameter estimates are unavailable, we are effectively guaranteed to have well-behaved estimates if we estimate using the MLE or Bayesian Posterior, and we have sufficient data.
- Much of the statistical theory of the 19/20th centuries was focused on finding UMVUE estimators. This includes our discussion on consistency and asymptotic efficiency.
- Unbiasedness is nice feature, but strictly requiring unbiasedness sometimes results in a few issues (Chapter 5, Pawitan, 2001).

- Finding an unbiased estimate can be extremely difficult.
- Not all parameter *have* an unbiased estimator.
- Requiring unbiasedness can produce terrible estimates.
- Regardless of the prior and likelihood, all estimates using the posterior means are biased (Section 7.5.2, Casella and Berger, 2024).
- The bias of good estimators (like the MLE) typically shrinks at a rate $O(n)$, where the variance typically shrinks at a rate $O(n^{1/2})$ (Chapter 5, Pawitan, 2001), meaning the variance usually is the cause for poor estimates, not the bias.

5 Sufficiency

Sufficiency

- One desirable trait of an estimator θ^* is that it it captures all of the information available in the data X about the parameter θ.
- Example: estimating $E[X_i] = \mu$, using: $\theta_1^* = X_1$, vs $\theta_2^* = \bar{X}_n$.
- We argued that both θ_1^* and θ_2^* are unbiased estimators, and that $\text{Var}(\theta_2^*) < \text{Var}(\theta_1^*)$. An additional way of viewing this, however, is that θ_1^* is *not capturing all of the information available for* θ
- This idea is known as the principle of *sufficiency*. This will be formally defined on the next slide.

Definition: Sufficiency
A statistic $T(X) = T(X_1, \ldots, X_n)$ is said to be *sufficient* for θ if the conditional distribution of $X_1, \ldots, X_n$, given $T = t$, does not depend on θ for any value of t.

- If we take some statistic sufficient T (could be an estimator), then after accounting for T, there is no more information available about θ in the available data.
- Formally, a sufficient statistic suggests that we could throw away all of the data, except the sufficient statistic, and not lose any information about the parameter θ.

Binomial sufficient statistic
Let $X_1, \ldots, X_n$ be iid Bernoulli random variables with parameter θ. Show that $T(X) = \sum_i X_i$ is a *sufficient statistic* for θ.

Proof. $T(X)$ is the number of "successes", and therefore $T(X)$ follows a Binomial(n, θ) distribution. We want to calculate the conditional probability function of $X = (X_1, \ldots, X_n)$, conditioned on $T(X)$. Both are discrete random variables, so we calculate the pmf as:

$$\begin{aligned}
f_{X|T_X}(x_1, \ldots, x_n | T(X) = t) &= \frac{\prod_{i=1}^n \theta^{x_i}(1-\theta)^{1-x_i}}{\binom{n}{t}\theta^t(1-\theta)^{n-t}} \\
&= \frac{\theta^{\sum_i x_i}(1-\theta)^{\sum_i (1-x_i)}}{\binom{n}{t}\theta^t(1-\theta)^{n-t}} \\
&= \frac{\theta^t(1-\theta)^{n-t}}{\binom{n}{t}\theta^t(1-\theta)^{n-t}} \\
&= \frac{1}{\binom{n}{t}}.
\end{aligned}$$

Note that this result does not contain θ. That means, conditioned on $T(X) = t$, the remaining randomness in the data X has no relation to the parameter θ. That is, the total number of 1s in this sample contains all of the information about θ that is available in the data. After accounting for this, other features (such as the exact value of X_3) contains no information about θ.
In short, $T(X)$ is a sufficient statistic for θ. □

Normal sufficient statistic
Let $X = X_1, \ldots, X_n$ be iid $N(\mu, \sigma^2)$, where σ^2 is known. Show that the mean, $T(X) = \bar{X}_n$ is a sufficient statistic for μ.

Proof. First, recall that we showed last semester that $\bar{X}_n \sim N\Big(\mu, \frac{\sigma^2}{n}\Big)$. Thus,

$$f_{T(X)}(t) = \frac{1}{\sqrt{2\pi\sigma^2/n}} \exp\Big(- \frac{n}{2\sigma^2}(t-\mu)^2)\Big).$$

Now, we want to find the pdf that corresponds to the conditional distribution $X|T(X)$. To do this, we will first simplify the joint distribution of X:

$$\begin{aligned} f_X(x_1, \ldots, x_n) &= \prod_{i=1}^n (2\pi\sigma^2)^{-1/2} \exp\Big(- \frac{1}{2\sigma^2}(x_i - \mu)^2\Big) \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(- \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2)\Big) \end{aligned}$$

In order to simplify the conditional distribution, we will introduce $t = \sum_i x_i/n$ into the equation:

$$\begin{aligned} f_X(x_1, \ldots, x_n) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(- \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - t + t - \mu)^2)\Big) \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(- \frac{1}{2\sigma^2} \big[\sum_{i=1}^n (x_i - t)^2 + n(t-\mu)^2\big]\Big), \end{aligned}$$

where the last expression results because the cross-product term involves $\sum_i (x_i - t) = 0$. □

With this, we can now compute the pdf of the conditional distribution $X|T(X)$:

$$\begin{aligned} f_{X|T(X)}(x|t) &= \frac{f_X(x)}{f_{T(X)}(t)} \\ &= \frac{\sqrt{2\pi\sigma^2/n}}{\sqrt{2\pi\sigma^2}} \frac{\exp\Big(- \frac{1}{2\sigma^2}\big[\sum_{i=1}^n (x_i - t)^2 + n(t-\mu)^2\big]\Big)}{\exp\Big(- \frac{n}{2\sigma^2}(t-\mu)^2)\Big)} \\ &= n^{-1/2}(2\pi\sigma^2)^{-(n-1)/2} \exp\Big(- \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - t)^2\Big). \end{aligned}$$

This conditional distribution does not involve μ, showing that $T(X) = \bar{X}_n$ is a sufficient statistic for μ.

- Using the definition to find a sufficient statistic is inconvenient. It requires knowing a candidate $T(X)$ ahead of time, knowing the distribution of the statistic, and computing conditional pmf / pdf.
- The following theorem provides us a way of finding sufficient statistics.

The Factorization Theorem
Let $f_X(x;\theta)$ be the joint pdf or pmf of a sample X. A statistic $T(X)$ is a sufficient statistic for θ if and only if there exist functions $g(t;\theta)$ and $h(x)$ such that, for all sample points x and parameter values θ,

$$f_X(x;\theta) = g\big(T(X);\theta\big)h(x).$$

The proof will only be for discrete distributions, though the proof for continuous distributions is similar. First, we'll suppose that $T(X)$ is a sufficient statistic. Let $g(t;\theta) = P_\theta(T(X) = t)$, and $h(x) = P(X = x|T(X) = T(x))$. Note that, by definition of sufficiency, the conditional distribution $h(x)$ does not depend on θ. Thus, we have:

$$\begin{aligned} f_X(x;\theta) &= P_\theta(X = x) \\ &= P_\theta\big(X = x \text{ and } T(X) = t(x)\big) \\ &= P\big(X = x|T(X) = t(x)\big)P_\theta\big(T(X) = t(x)\big) \\ &= g\big(T(X);\theta\big)h(x), \end{aligned}$$

Giving the desired factorization.
We now assume that the factorization exists. Let $q(t;\theta)$ be the pmf of $T(X)$. Note that there are many possible sets for X such that $T(X) = t$ (e.g., if $T(X)$ is the mean, then $X = (-1, 1)$, or $X = (1, -1)$, or $X = (0, 0)$ all give the same value $T(X) = 0$.)
For a fixed value t, define $A_t = \{x : T(x) = t\}$. Then, the PMF of $X|T(X) = t$ is:

$$\frac{f_X(x;\theta)}{q\big(T(x);\theta\big)} = \frac{g\big(T(x);\theta\big)h(x)}{q\big(T(x);\theta\big)}.$$

Using the variable transformation theorem for discrete random variables, we have

$$\begin{aligned} q\big(T(x);\theta\big) &= \sum_{y\in A_{T(x)}} f_Y(y;\theta) \\ &= \sum_{y\in A_{T(x)}} g\big(T(y);\theta\big)h(y) \\ &= g\big(T(x)\big) \sum_{y\in A_{T(x)}} h(y), \end{aligned}$$

where the last equality comes from the definition of the set $A_{T(x)}$, since $y \in A_{T(x)}$ implies that $T(y) = T(x)$. Therefore, the conditional distribution is:

$$\begin{aligned} \frac{f_X(x;\theta)}{q\big(T(x);\theta\big)} &= \frac{g\big(T(x);\theta\big)h(x)}{q\big(T(x);\theta\big)} \\ &= \frac{g\big(T(x);\theta\big)h(x)}{g\big(T(x)\big)\sum_{y\in A_{T(x)}} h(y)} \\ &= \frac{h(x)}{\sum_{y\in A_{T(x)}} h(y)}. \end{aligned}$$

By assumption, $h(x)$ does not involve the parameter θ, proving that if we can factor the joint pmf, then the conditional distribution $X|T(X)$ does not include θ, or that $T(X)$ is a sufficient statistic for θ.

- The factorization theorem gives us a way to find sufficient statistics:
- Factor the joint pdf / pmf into two parts: one depending on θ, and one part that does not depend on θ. The part without θ becomes the function $h(x)$. In the remaining part of the function, find the remaining function of the data $T(X)$.

Example: Normal (known variance)
Let $X_i \overset{iid}{\sim} N(\mu, \sigma^2)$, with σ^2 known. Find a sufficient statistic for μ.

Proof. We already showed that $T(X) = \bar{X}_n$ is a sufficient statistic for μ. Suppose that we weren't given $T(X)$, but had to instead find it. We can use the factorization theorem to do this. The joint pdf is given by:

$$\begin{aligned} f_X(x_1, \ldots, x_n) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{1}{2\sigma^2} \sum_{i=1}^n \big(x_i - \bar{x} + \bar{x} - \mu)^2\big)\Big) \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{1}{2\sigma^2} \big[\sum_{i=1}^n \big(x_i - \bar{x})^2 + n(\bar{x} - \mu)^2\big]\Big) \\ &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{1}{2\sigma^2} \sum_{i=1}^n \big(x_i - \bar{x})^2\Big) \exp\Big(-\frac{n}{2\sigma^2}(\bar{x} - \mu)^2\Big) \end{aligned}$$

In the expression above, we can let:

$$h(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{1}{2\sigma^2} \sum_{i=1}^n \big(x_i - \bar{x})^2\Big),$$

which is only a function of the data, and not of the parameter μ. This gives us the remaining function $g\big(T(x); \mu\big)$:

$$g\big(T(x); \mu\big) = \exp\Big(-\frac{n}{2\sigma^2}(\bar{x} - \mu)^2\Big).$$

Here, we can find what the sufficient statistic is. In particular, functions of the data in this function are sufficient statistics, by the factorization theorem. Therefore: $T(X) = \bar{X}$ is a sufficient statistic for μ. □

- The Factorization Theorem gives us a way to find sufficient statistics, but these statistics are not unique.
- For instance, if $T(X)$ is a sufficient statistic, than $cT(X)$ is also sufficient (for any constant c).
- Furthermore, the factorization theorem states that the entire vector of data X is itself a sufficient statistic:
$$f_X(x; \theta) = f_X(x; \theta)h(x),$$
where $h(x) = 1$. This satisfies all conditions of the Factorization Theorem, so $T(X) = X$ is sufficient for θ.
- This leads us to the idea of *minimal sufficiency*.

Definition: Minimal Sufficient Statistic
A sufficient statistic $T(X)$ is called a *minimal sufficient statistic* if, for any other sufficient statistic $T'(X)$, $T(x)$ is is a function of $T'(x)$.

- One way of thinking about this is that a minimal sufficient statistic is the *coarsest* possible partition for a sufficient statistic, and thus achieves the greatest possible data reduction for a sufficient statistic.
- In the normal model, for instance, both $T(X) = \bar{X}$ and $T'(X) = X$ were shown to be sufficient. $T'(X) = X$ *cannot* be minimally sufficient, because it cannot be recovered from the mean alone (i.e., X is not a function of $\bar{X}$).

- Finding a minimally sufficient statistic can be difficult, but there are theorems that can be used to help. See, for instance, Theorem 6.2.13 from Casella and Berger (2024).
- We'll finish discussion about sufficieny with one last example.

Joint sufficiency: IID normal model
Let X_i be iid $N(\mu, \sigma^2)$, where both μ and σ^2 are unknown. Find a sufficient statistic for $\theta = (\mu, \sigma^2)$. First, consider the joint density:

$$\begin{aligned} f(x_1, \ldots, x_n; \mu, \sigma^2) &= \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} \exp\Big[-\frac{1}{2\sigma^2}(x_i - \mu)^2 \Big] \\ &= \frac{1}{\sigma^n (2\pi)^{n/2}} \exp\Big[-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2 \Big] \\ &= \frac{1}{\sigma^n (2\pi)^{n/2}} \exp\Big[-\frac{1}{2\sigma^2} \Big(\sum_{i=1}^{n} x_i^2 - 2\mu \sum_{i=1}^{n} x_i + n\mu^2 \Big) \Big]. \end{aligned}$$

In the final expression above, we have factored the joint pdf into an expression that involves two statistics: $T_1(x) = \sum_i x_i$, and $T_2(x) = \sum_{i=1} x_i^2$. Using the factorization theorem, we let $h(x) = 1$, and:

$$g\big(T_1(x), T_2(x); \mu, \sigma^2\big) = \frac{1}{\sigma^n (2\pi)^{n/2}} \exp\Big[-\frac{1}{2\sigma^2} \Big(T_2(x) - 2\mu T_1(x) + n\mu^2 \Big) \Big],$$

showing that $\Big(\sum_i x_i, \sum_i x_i^2 \Big)$ is a sufficient statistic for $\theta = (\mu, \sigma^2)$. We will not prove it here, but this actually is a minimal sufficient statistic for θ.
In this example (and all previous examples), we found that the dimension of a minimal sufficient statistic matched the dimensions of the parameter vector θ. The study of when this occurs led to the development of what is called the *exponential family* of probability distributions. This is a really important family because it has very nice properties. This family includes Gaussian, Poisson, binomial, and Gamma distributions.

Rao-Blackwell Theorem

- In addition to having desireable proprties, sufficient statistics can be used to improve estimators. This is known as *Rao-Blackwellization*.

The Rao-Blackwell Theorem
TODO: Finish this theorem

- TODO: (page 310, Rice: Rao-Blackwell Theorem states we should use sufficient statistics when available)
- TODO: The MLE is a function of sufficient statistics (page 309 Rice, Corollary A)
- TODO: Theorem 3.2, Pawitan (p 56): The likelihood function is a minimal sufficient statistic.

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